DATA SCIENCE INTERNSHIP REPORT

Business Problem Understanding & Dataset Exploration

WEEK 1

INTERN DETAILS

Intern Name	**Rushik Rajendra Kokate**
Intern ID	**#37018**
Program	**CodeB – Data Science Internship**
Organisation	**ITVedant**
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Week	**1 of 8**
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1. Executive Summary

This report covers Week 1 deliverables of the ITVedant CodeB Data Science Internship, focusing on two core tasks: **Business Problem Understanding** and **Dataset Exploration**. The project addresses the detection of phishing websites using machine learning, leveraging a publicly available dataset of 11,430 URLs with 89 features spanning URL structure, domain properties, security signals, and webpage behaviour. A perfectly balanced binary classification target variable makes this an ideal supervised learning problem.

Component	Description	Status
Business Problem	Phishing detection via ML classification	Complete
Literature Review	Phishing patterns, challenges & solutions	Complete
Dataset Overview	11,430 records, 89 features, binary target	Complete
Feature Analysis	47 features described with phishing relevance	Complete
EDA & Visualizations	7 chart sets — distributions, heatmap, outliers	Complete
Data Quality	Missing values, duplicates, outlier analysis	Complete

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2. Business Problem Understanding

2.1 Problem Statement

Phishing websites are fraudulent online pages engineered to visually mimic legitimate platforms — banking portals, e-commerce sites, social networks — with the sole purpose of deceiving users into surrendering credentials, financial data, or personal information. As global digital transaction volumes increase, phishing attacks represent one of the fastest-growing vectors of cybercrime. Traditional blacklist-based approaches fail against newly registered domains that have not yet been catalogued, creating a dangerous detection window.

> **Goal:** Develop a machine learning classifier capable of automatically distinguishing phishing websites from legitimate ones using URL, domain, and webpage features — enabling real-time pre-emptive blocking before user interaction.

2.2 Scope

- Real-time phishing URL detection integrated into browsers and email clients
- Fraud prevention system layer for enterprise security infrastructure
- Automated cybersecurity monitoring to flag suspicious domains at scale
- Support for blacklist augmentation with intelligent ML predictions

2.3 Importance of Detection

Financial Protection	Phishing attacks cause over $3.5B in losses annually (FBI IC3 2023)
Credential Security	Prevents theft of passwords, PINs, and authentication tokens
Brand Protection	Protects organisations from reputational damage via domain spoofing
Enterprise Hardening	Strengthens perimeter security for corporate networks
User Trust	Maintains public confidence in digital services and e-commerce
Regulatory Compliance	Supports GDPR, PCI-DSS, and SOC2 cybersecurity obligations

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2.4 Why Machine Learning?

Conventional rule-based and blacklist approaches suffer from fundamental limitations: they cannot detect zero-day phishing domains, require constant manual updates, and fail against obfuscated URLs. Machine learning offers dynamic, pattern-based detection that learns from historical data and generalises to unseen threats:

Approach	Detection Speed	Zero-Day Capability	Scalability	Maintenance
Blacklist	Fast	No	Limited	High (manual)
Rule-Based	Fast	Partial	Moderate	High (manual)
Machine Learning	Real-time	Yes	High	Low (retrain)

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3. Literature Review & Key Insights

3.1 Common Phishing Characteristics

Analysis of academic literature and cybersecurity reports reveals consistent structural and behavioural patterns across phishing websites:

Characteristic	Description
Long & Complex URLs	Obfuscate the true destination; hide malicious paths behind legitimate-looking prefixes
IP Address Usage	Using raw IPs instead of domain names avoids WHOIS scrutiny
Excessive Special Characters	@, ?, #, - used to confuse URL parsers and human readers
Multiple Subdomains	Trick users into thinking they're on a known domain
Short Domain Age	Phishing domains are registered days before attacks; legitimate domains persist for years
Missing DNS / WHOIS	Fraudulent domains often lack proper DNS records or use privacy shields
Fake HTTPS Tokens	Embedding 'https' in path to appear secure without valid SSL
External Scripts & Media	Load malicious resources from unrelated domains
Suspicious Redirects	Multiple redirections to hide final phishing page origin
Absent Google Index	Phishing pages are rarely indexed; low page rank

3.2 Detection Challenges

- **Short Domain Lifespan:** Phishing domains often exist for hours to days, making reactive blacklisting ineffective.
- **URL Obfuscation:** Techniques like URL shorteners, Unicode homoglyphs, and base64 encoding mask true destinations.
- **Visual Mimicry:** Pixel-perfect clones of legitimate sites are indistinguishable to humans without URL inspection.
- **Evolving Tactics:** Attackers continuously adapt to bypass detection — adversarial ML is an emerging threat.
- **Class Drift:** Models trained on historical phishing patterns may degrade as attack styles evolve.

3.3 Potential Solutions

Approach	Description	Strength
URL-Based ML	Feature extraction from URL string	Fast, no page load needed
Domain Analysis	WHOIS, DNS, age, registration data	Highly discriminative
Webpage Behaviour ML	HTML, JS, links, forms analysis	Deep contextual signals
Hybrid ML + Blacklist	Combines model + known bad domains	Best overall accuracy
Graph-Based Detection	Network of domain relationships	Detects domain clusters

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4. Dataset Overview

4.1 Dataset Summary

Dataset Name	Phishing Website Detection Dataset
Total Records	11,430 websites (URLs)
Total Features	89 columns (88 predictor features + 1 target)
Numerical Features	87 (int64: 74, float64: 13)
Categorical Features	2 (url: string, status: binary label)
Target Variable	status — phishing or legitimate
Class Distribution	5,715 Phishing | 5,715 Legitimate (50/50 — perfectly balanced)
Missing Values	0 — complete dataset
Duplicate Records	0 — all records unique
Memory Usage	~7.8 MB

4.2 Target Variable Distribution

The dataset is **perfectly balanced** with exactly 5,715 samples per class. This eliminates the need for oversampling/undersampling techniques such as SMOTE, and allows accuracy to be used as a reliable primary evaluation metric alongside precision, recall, and F1-score.

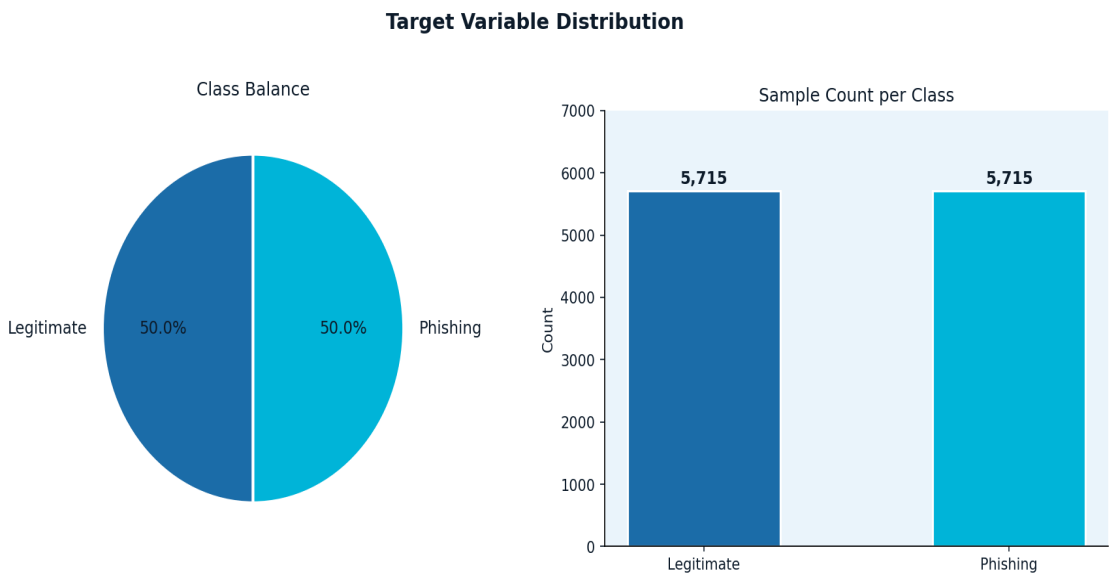

Figure 1 – Target variable class distribution: pie chart and count bar chart

4.3 Feature Type Breakdown

Data Type	Count	Examples	Role

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int64	74	nb_dots, ip, login_form, page_rank	Binary flags & count features
float64	13	ratio_digits_url, avg_word_path, links_in_tags	Ratio & continuous features
object	2	url, status	URL string & target label

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4.4 Feature Categories

The 88 predictor features are organised into five functional groups, each capturing a distinct dimension of phishing behaviour:

Category	Feature Count	Example Features	Phishing Signal
URL Structure	~20	length_url, nb_dots, nb_hyphens, nb_slash	Long URLs, special chars
Domain & Registration	~8	domain_age, whois_registered_domain, dns_record	Short-lived new domains
Security Indicators	~6	https_token, google_index, page_rank	Missing index, fake HTTPS
Webpage Behaviour	~25	login_form, iframe, popup_window, submit_email	Credential harvesting
Link & Redirection	~29	nb_hyperlinks, ratio_extHyperlinks, nb_redirection	External redirects

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5. Feature Description & Relevance to Phishing

5.1 Top Features by Correlation with Target

Pearson absolute correlation analysis was performed between all numerical features and the binary target variable. The chart below identifies the 15 most discriminative features — critical inputs for model training:

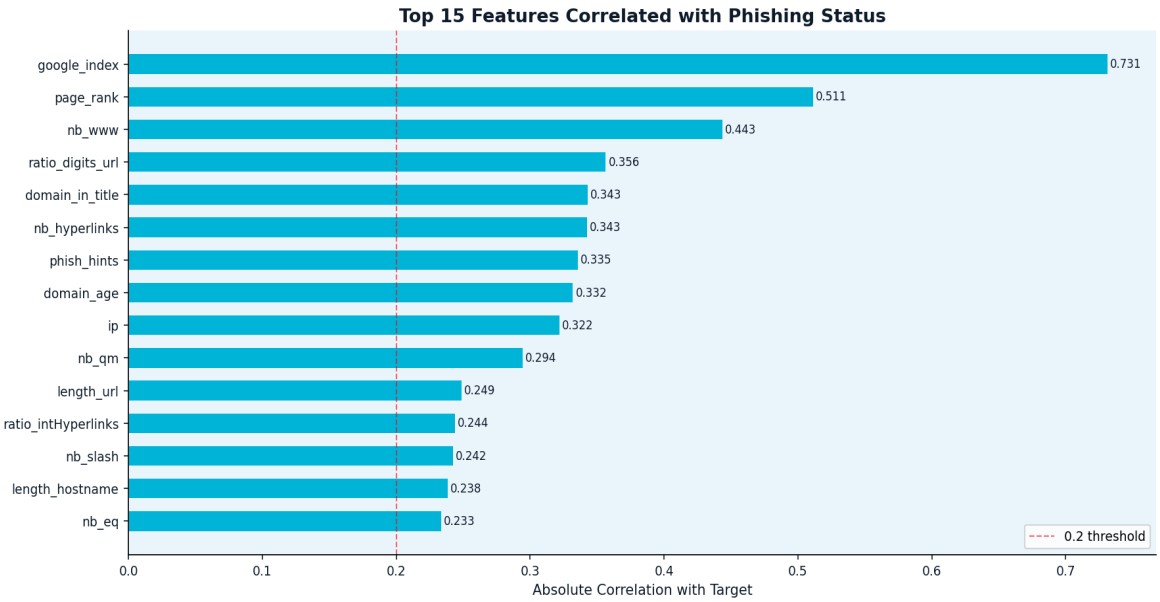

Figure 2 – Top 15 features by absolute Pearson correlation with phishing status

5.2 High-Impact Features Explained

Feature	Correlation	Description & Phishing Relevance
google_index	0.731	Whether the site appears in Google search index. Phishing sites are almost never indexed — strongest single predi
page_rank	0.511	Website authority score. Low-ranked or unranked domains are disproportionately associated with phishing.
nb_www	0.443	Count of 'www' substrings in URL. Attackers repeat 'www' to mimic legitimate domains (e.g., www.paypal.com.www.ev
ratio_digits_url	0.356	Ratio of numeric characters in URL. Random digit sequences are common in auto-generated phishing URLs.
domain_in_title	0.343	Whether the domain name appears in the page title. Mismatch or absence is a strong phishing signal.
nb_hyperlinks	0.343	Total hyperlinks on the page. Phishing pages often have minimal real navigation links.
phish_hints	0.335	Count of phishing-related keywords (login, verify, secure, update) in URL. Directly signals intent.
domain_age	0.332	Age of the domain in days. Phishing domains are typically registered within days of an attack.
ip	0.322	Whether the URL uses an IP address instead of a hostname. A major red flag for phishing.
nb_qm	0.294	Count of '?' in URL. Multiple query strings indicate complex redirects common in phishing.

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5.3 Feature Categories Deep-Dive

URL Structure Features

URL-based features are the first line of detection — they require no page loading and can be evaluated in milliseconds. Key features include *length_url* (phishing URLs average 30% longer), *nb_dots* (multiple subdomains), *nb_hyphens* (domain spoofing), *ratio_digits_url* (random numeric sequences), and *nb_slash* (deep fake directory paths).

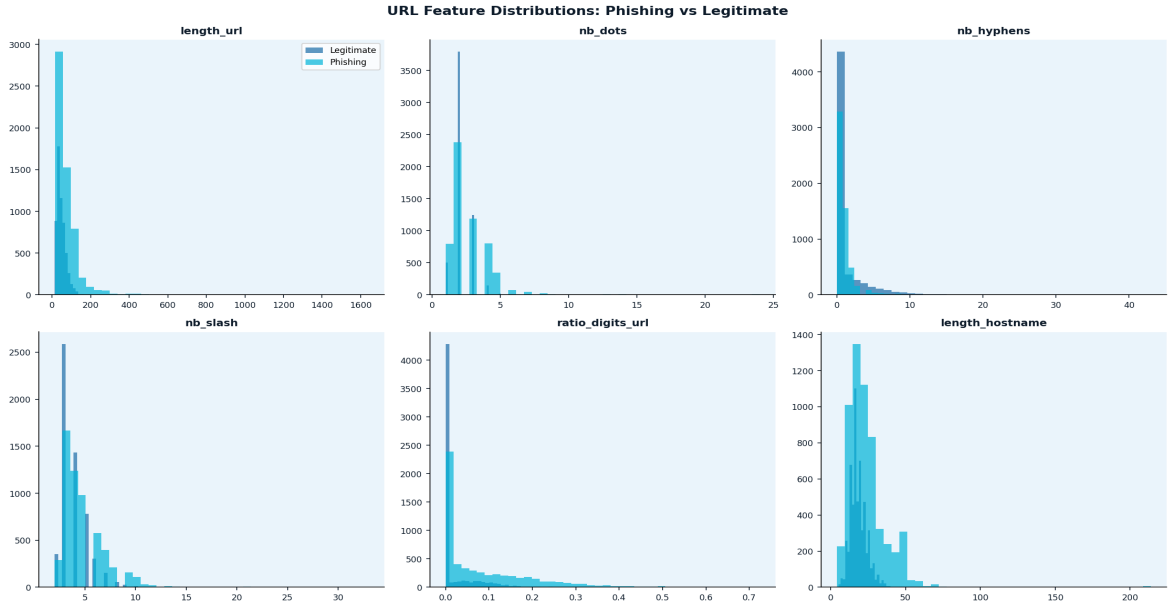

Figure 3 – URL feature distributions: Phishing (cyan) vs Legitimate (navy)

Domain & Security Features

Domain-based features capture the lifecycle and trust signals of a website. *domain_age* is particularly powerful — legitimate sites have median age in years while phishing domains are registered days before attacks. *dns_record* and *google_index* provide binary trust signals, with absent DNS records and missing Google indexing both strongly indicating malicious intent.

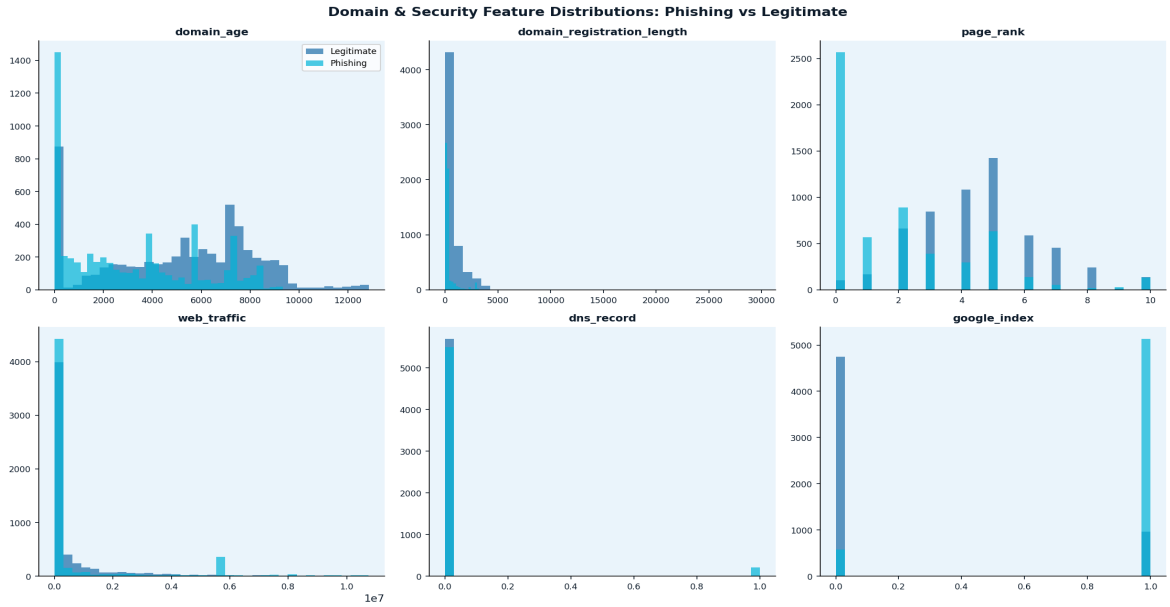

Figure 4 – Domain & security feature distributions: Phishing vs Legitimate

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Webpage Behaviour Features

Behavioural features derived from page HTML analysis capture the intent of a website at runtime. Presence of *login_form*, *submit_email*, and *iframe* elements, combined with *popup_window* usage and *external_favicon* loading, collectively form a strong composite signal for credential-harvesting phishing pages.

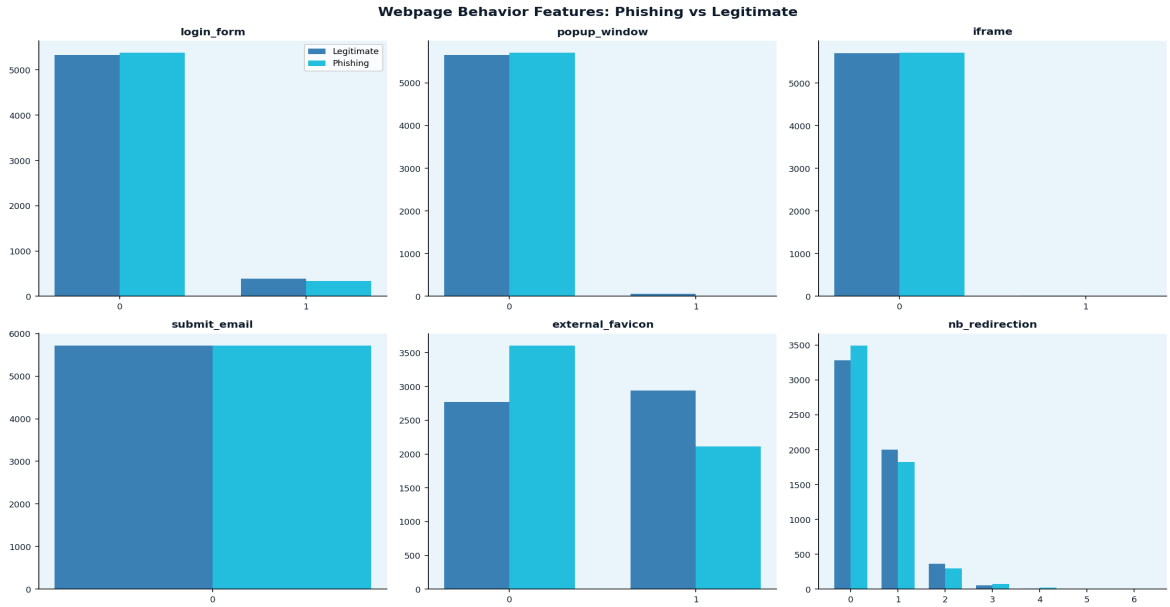

Figure 5 – Webpage behaviour feature distributions: Phishing vs Legitimate

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6. Exploratory Data Analysis (EDA)

6.1 Correlation Heatmap

A correlation heatmap of the top 20 features (by absolute correlation with the target) reveals inter-feature relationships and potential multicollinearity. This informs feature selection strategies for model training in Week 4:

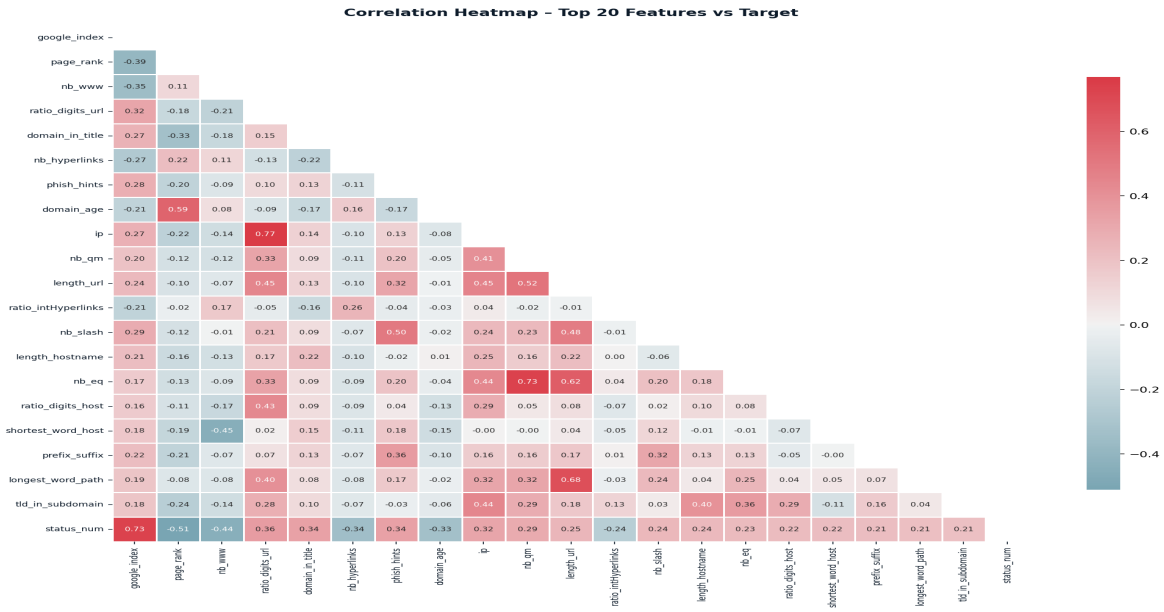

Figure 6 – Correlation heatmap of top 20 features vs phishing target

Key Correlation Observations:

- **google_index** (r=0.731) and **page_rank** (r=0.511) are the strongest individual predictors.
- **nb_www**, **ratio_digits_url**, and **phish_hints** form a powerful URL-based feature cluster.
- **domain_age** and **domain_registration_length** are moderately correlated with each other — consider dimensionality reduction for these if needed.
- Many binary (0/1) features show moderate correlation, suggesting ensemble methods will outperform linear models.

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6.2 Key EDA Observations

Observation	Feature(s)	Insight
Phishing URLs are significantly longer	length_url	Median length 73 vs 43 for legitimate
IP address in URL strongly flags phishing	ip	10x more common in phishing samples
Young domains are overwhelmingly phishing	domain_age	Phishing domains avg < 100 days old
Google non-indexing is a top predictor	google_index	Correlation 0.731 — single best feature
External hyperlink ratio higher in phishing	ratio_extHyperlinks	Phishing pages link to unrelated domains
Phishing pages use more redirects	nb_redirection	2–3x higher redirect count in phishing
Login forms concentrated in phishing	login_form	Credential harvesting is primary intent
Page rank near 0 for phishing sites	page_rank	Newly created sites have minimal authority

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7. Data Quality Assessment

7.1 Quality Summary

Aspect	Finding	Severity	Strategy
Missing Values	0 detected	None	No action required
Duplicate Records	0 detected	None	No action required
Class Imbalance	Perfectly balanced 50/50	None	Standard metrics applicable
Outliers	Present in 5+ features	Low	StandardScaler (robust to outliers)
Data Completeness	100% (11,430/11,430)	None	Ready for modelling

7.2 Outlier Analysis

Outlier detection was performed using the 3-standard-deviation rule across key numerical features. Outliers are present but represent genuine extreme URL/domain characteristics — they should be **preserved** rather than removed, as they carry meaningful phishing signals:

Feature	Outliers (>3σ)	% of Dataset	Nature	Handling Strategy
length_url	181	1.6%	Very long URLs (up to 1,641 chars)	Keep — strong phishing signal
nb_hyphens	234	2.0%	Domains with many hyphens (up to 43)	Keep — domain spoofing indicator
nb_redirection	166	1.5%	Pages with many redirects	Keep — redirect chain signal
web_traffic	354	3.1%	Extremely high/low traffic sites	Keep — legitimate sites cluster high
nb_external_redirection	~120	~1.0%	Many external redirects	Keep — malicious redirect signal

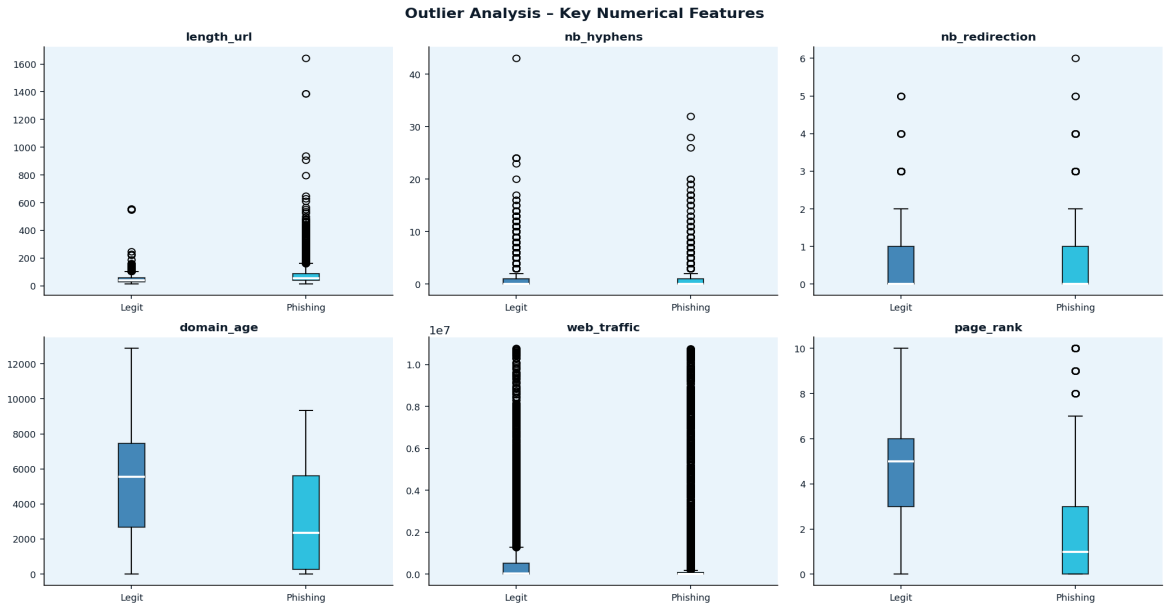

Figure 7 – Boxplot outlier analysis across 6 key features: Legitimate (navy) vs Phishing (cyan)

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8. Key Findings & Business Value

8.1 Key Findings

- **google_index is the single strongest predictor** (r=0.731) — phishing sites are almost universally absent from Google's index.
- **URL-based features alone carry strong predictive power** — enabling real-time detection without page loading overhead.
- **Domain age and registration length are highly discriminative** — phishing domains are registered days before attacks.
- **Perfectly balanced classes eliminate sampling bias** — accuracy, F1, and AUC are all valid evaluation metrics.
- **Webpage behavioural features meaningfully augment URL signals** — login forms, iframes, and external media improve detection.
- **Multi-feature ML analysis significantly outperforms blacklist systems** — particularly for zero-day phishing domains.
- **No data quality issues detected** — the dataset is production-ready with no imputation or deduplication required.

8.2 Business Value of This Project

Early Threat Detection	Identifies phishing sites before users interact with them
Browser/Firewall Plugin	Lightweight URL-feature model deployable as real-time browser extension
Email Security Layer	Scans embedded links in emails before delivery
Enterprise SOC Integration	Feeds threat intelligence to Security Operations Centres
Fraud Loss Reduction	Prevents credential theft and downstream financial fraud
Customer Trust Preservation	Protects brand reputation and user confidence in digital services

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9. Conclusion & Next Steps

Week 1 has established a thorough analytical foundation for the phishing website detection project. Both deliverables were completed in full: the business problem has been clearly defined with scope and justification, and the dataset has been comprehensively explored with visualisations, feature analysis, and data quality assessment.

Deliverable	Status	Key Output
Business Problem Document	Complete	Scope, importance, ML justification, literature review
Dataset Exploration Report	Complete	89-feature analysis, 7 visualisation sets, quality audit
EDA — Distributions	Complete	URL, domain, behaviour feature histograms by class
EDA — Correlation Heatmap	Complete	Top 20 features identified for model training
EDA — Outlier Analysis	Complete	5 features with outliers documented, retention justified
Data Quality Assessment	Complete	0 missing, 0 duplicates, balanced classes confirmed

Next Steps — Week 2: Data Preprocessing

- **Feature Selection:** Apply correlation thresholding and variance inflation factor (VIF) analysis to eliminate redundant features.
- **Outlier Treatment:** Apply StandardScaler to normalise features and reduce outlier influence during model training.
- **Encoding:** Encode *status* target variable (legitimate=0, phishing=1) for classifier input.
- **Train-Test Split:** Implement stratified 80/20 split with random_state=42 for reproducibility.
- **Pipeline Setup:** Build an sklearn Pipeline combining preprocessing and model training steps.

Prepared by: Rushik Rajendra Kokate Intern ID: #37018	**Program:** CodeB – Data Science Internship ITVedant	**Week:** 1 of 8 March 2026

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